

BIRLA INSTITUTE OF TECHNOLOGY MESRA, RANCHI
DEPARTMENT OF MATHEMATICS

MA24103 Mathematics-II

Session: SP 2026

Tutorial - 3 (Module III:Fourier Series)

1. Find the Fourier series of $f(x) = \pi^2 - x^2$ in $-\pi < x < \pi$.
2. Find the Fourier series for the function $f(x)$ if $f(x)$ is defined in $-\pi < x < \pi$ as

$$f(x) = \begin{cases} 0 & -\pi < x < 0 \\ \sin x & 0 < x < \pi \end{cases}$$

Hence deduce that

$$\frac{1}{4}(\pi - 2) = \frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots$$

3. Obtain a Fourier expression for $\sqrt{1 - \cos x}$ in the interval $-\pi < x < \pi$.
4. Find the Fourier series for the function

$$f(x) = \begin{cases} -h & -\pi < x < 0 \\ h & 0 < x < \pi \end{cases}$$

here, $h \neq 0$. Hence deduce that

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

5. Consider the function $f(x) = e^x$ in $0 < x < 2$.
 1. Determine the Fourier series of $f(x)$.
 2. Determine the half-range sine series of $f(x)$.
 3. Determine the half-range cosine series of $f(x)$.
6. Determine the half-range Fourier sine series for $f(x) = x(\pi - x)$ in $0 < x < \pi$. Hence deduce

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)^3} = \frac{\pi^3}{32}$$

7. Determine the half-range sine and cosine series for $f(x) = \sin(x)$, where $0 < x < \pi$.
8. Determine the half-range Fourier cosine series for $f(x) = \sin(\frac{\pi x}{l})$, where $0 < x < l$.
9. Obtain the half range Fourier cosine and sine series for

$$\begin{cases} 0 & 0 < x < \frac{\pi}{2} \\ \pi - x & \frac{\pi}{2} < x < \pi \end{cases}$$